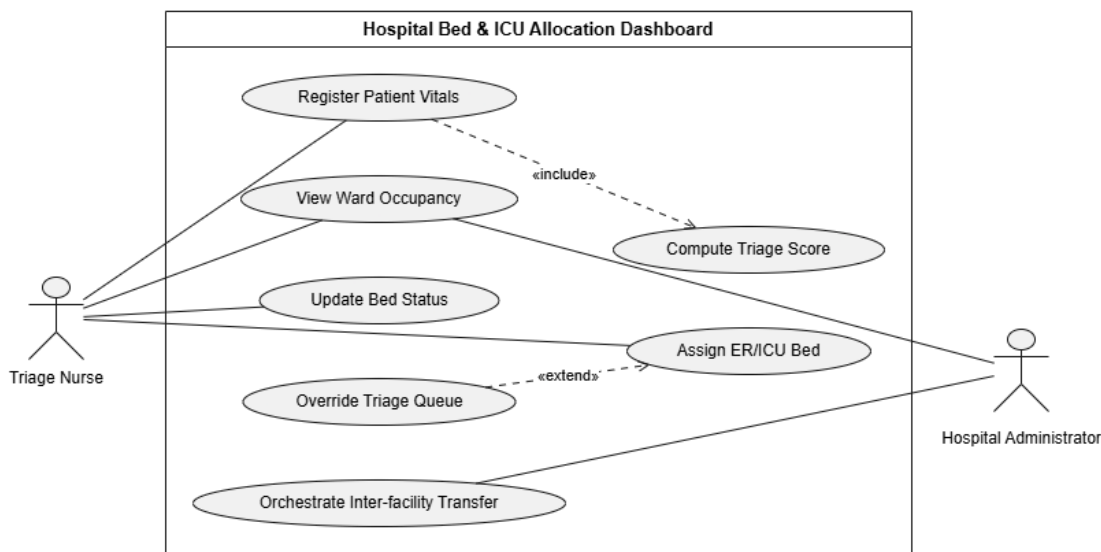


1. Requirements Table:

ID	Type / Priority	Description	Acceptance Criteria	Rationale
FR-001	High	The system shall compute triage severity scores from intake vitals and dynamically order the ER bed allocation waitlist.	Pass: Higher acuity patient automatically moves ahead of lower acuity patient. Fail: FIFO queue ignores critical emergency.	Ensures the most critical patients receive care first, adhering to medical triage protocols.
FR-002	High	The system shall display a real-time dashboard of all ER and ICU bed statuses (Available, Occupied, Cleaning, Maintenance).	Pass: Dashboard reflects status changes instantly without manual page refresh. Fail: Staff sees outdated bed availability.	Allows administrators and nurses to make immediate routing decisions based on accurate capacity.
FR-003	High	The system shall allow the Triage Nurse to assign a waitlisted patient to an "Available" ER or ICU bed.	Pass: Patient is removed from waitlist and linked to the specific bed; bed status changes to "Occupied."	Core operational function for admitting patients and tracking who is where.
FR-004	Medium	The system shall allow authorized users to initiate an inter-facility ambulance transfer workflow for a patient if local ICU beds are full.	Pass: User can select a patient, select a destination facility, and generate a dispatch request.	Necessary for orchestration of care when local hospital capacity is exceeded.

ID	Type / Priority	Description	Acceptance Criteria	Rationale
FR-005	Medium	The system shall allow Hospital Administrators to generate daily ward occupancy and transfer volume reports.	Pass: Administrator can export a PDF/CSV showing peak occupancy times and transfer counts.	Required for capacity planning, staffing adjustments, and operational auditing.
NFR-001	Performance	Bed status transitions (Occupied, Cleaning, Available) must propagate to all connected dashboard clients in under 1 second via WebSockets.	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Real-time accuracy is critical in emergency settings; delays could cause double-booking of beds.
NFR-002	Security	The dashboard must enforce Role-Based Access Control (RBAC), ensuring only authorized clinical staff can view Patient Protected Health Information (PHI).	Pass: An administrator account cannot view patient intake vitals, while a triage nurse account can.	Strict compliance with HIPAA (or local healthcare data laws) is mandatory for hospital software.

2. UML Use-Case Diagram:



3. Use Case Flow Specification:

Use Case Name: Assign ER/ICU Bed **Primary Actor:** Triage Nurse **Goal:** To allocate an available bed to the highest-priority patient on the triage waitlist.

Preconditions:

- The Triage Nurse is authenticated and logged into the dashboard.
- The patient has been registered, vitals entered, and a triage severity score computed.
- The patient is currently on the active waitlist.

Postconditions:

- The patient is removed from the waitlist.
- The selected bed's status is updated to "Occupied".
- The real-time dashboard broadcasts the updated capacity to all connected clients.

Main Success Scenario:

1. The Triage Nurse views the "Dynamic ER Waitlist" on the dashboard.
2. The system displays the waitlist, ordered descending by Triage Severity Score.
3. The Triage Nurse selects the patient at the top of the queue.
4. The system highlights all beds currently marked as "Available" on the ward map.
5. The Triage Nurse clicks on an "Available" bed to assign the patient.
6. The system prompts for confirmation of the assignment.
7. The Triage Nurse confirms.
8. The system links the patient ID to the Bed ID, updates the bed status to "Occupied," and removes the patient from the waitlist.

Alternate Flow 1: No Available Beds (Triggering Inter-facility Transfer)

- *At Step 4:* The system determines there are zero beds marked "Available" in the required ward (e.g., ICU).
- 4a. The system displays an "ICU at Capacity" alert and grays out the ward map.

- 4b. The system prompts the Triage Nurse to initiate an "Inter-facility Ambulance Transfer."
- 4c. The Triage Nurse selects "Initiate Transfer."
- 4d. The system branches to the *Orchestrate Inter-facility Transfer* use case, alerting the Hospital Administrator and generating an ambulance dispatch request.